Name: _____ **Branch:** _____ **SNU ID:** _____

1. Which of the following is true?
(a) The AVL Trees are more balanced as compared to R-B Trees, but they may cause more rotations during insertions and deletions.
(b) Heights of AVL and R-B Trees are generally same, but AVL Trees may cause more rotations during insertions and deletions.
(c) R-B Trees are more balanced as compared to AVL Trees, but may cause more rotations during insertions and deletions.
(d) Heights of AVL and R-B Trees are generally same, but R-B Trees may cause more rotations during insertions and deletions.

2. What is the number of NULL links in a Binary Tree of 'n' nodes?
(a) n (b) n+1 (c) n-1 (d) n/2

3. Minimum numbers of nodes in AVL tree of height 5 is (with root at height = 1)
(a) 11 (b) 12 (c) 13 (d) 14

4. Which of the following is true about AVL and R-B Trees?
(a) In AVL tree, insert( ) operation, we first traverse from root to newly inserted node and then from newly inserted node to root. While in Red Black tree insert(), we only traverse once from root to newly inserted node.
(b) In both AVL and Red Black insert operations, we traverse only once from root to newly inserted node,
(c) In both AVL and Red Black insert operations, we traverse twice, first traverse root to newly inserted node and then from newly inserted node to root.
(d) None of the above

5. Is the following statement valid?

A Red-Black Tree which is also a perfect Binary Tree can have all black nodes.
(a) Yes (b) No

6. Which of the following data structures has the least height
(a) B-Tree of order 4
(b) B-Tree of order 3
(c) B-Tree of order 5
(d) B-Tree of order 6

7. Which of the following is true about the AVL Tree?
1. AVL Tree is a binary Tree
2. The search time of AVL tree will be always less than (or) equal to the search time of binary search tree
3. AVL tree is Binary Search Tree with maximum balancing factor of 2

(a) Only (2) (b) Only (1) and (2) (c) Only (1) and (3) (d) All of the above

8. In an M-way tree if degree of tree is M, then maximum values per node and maximum number of subtrees are
(a) M-1, M (b) M+1, M-1 (c) M, M (d) M-1, M-1

9. In threaded binary tree all right child NULL pointers point to
(a) Postorder successor of the node
(b) Inorder predecessor of the node
(c) Inorder successor of the node
(d) Preorder successor of the node

10. Construct an AVL Tree with the elements
21, 26, 30, 9, 4, 14, 28, 18

I. What will be the root node for the AVL tree?
(a) 26 (b) 21 (c) 28 (d) 18

II. What will be the postorder predecessor of the root node, in the above resultant tree?
(a) 26 (b) 21 (c) 28 (d) 9

11. Consider the following code:

```
struct binary{
  struct binary *left;
  int data;
  struct binary *right;
}

void F1(struct binary *t)
{
  if(t)
  {
    printf("%d", t->data);
    F2(t->left);
    F2(t->right);
  }
}

void F2(struct binary *t)
{
  if(t)
  {
    F1(t->left);
    printf("%d", t->data);
    F1(t->right);
  }
}
```

Now, consider the following tree

```
      A
     / \
    B   D
   /   /
  C   E
       \
        F
```

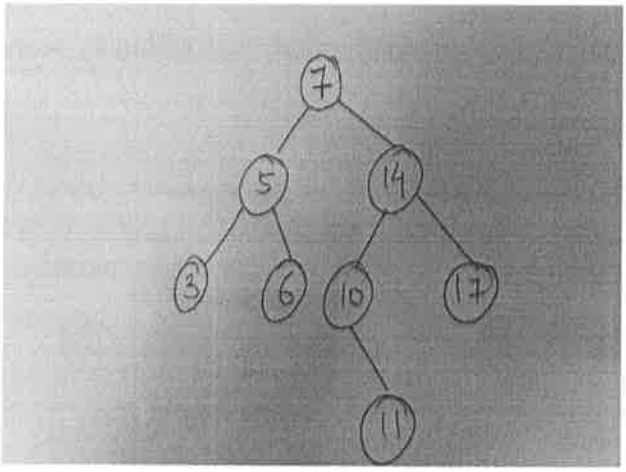
I. If F1(t) is called on the above tree with t as root node (A), what is its output?

(a) ABCDEF (b) ACBFED (c) ABDCFE (d) ACBEFD

II. When F2(t) is called on this tree, with root as the parameter that is (F2(root)), what is its output?

(a) BCADEF (b) BCDEFA (c) DEFBCA (d) DEFABC

12. Consider the given AVL-Tree



If the element with value '12' is inserted into this AVL tree, how many rotations are required to balance the tree?

(a) 0
(b) 1
(c) 2
(d) 3

13. In which order, the element must be inserted into an AVL tree, so that, no single rotation is required, for the given elements = { 1, 2, 3, 4, 5, 6, 7}

(a) { 4, 2, 6, 1, 3, 5, 7 }
(b) { 4, 2, 1, 3, 5, 6, 7 }
(c) { 7, 6, 5, 4, 2, 1, 3 }
(d) { 4, 1, 2, 3, 5, 6, 7 }

14. Consider the given code:

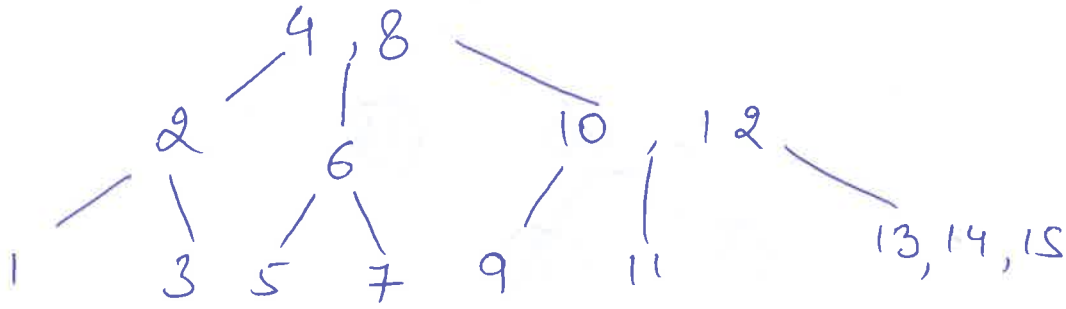
```
int fun(struct binarytreenode *root1, struct binarytreenode *root2)
{
if(root1 == NULL && root2 == NULL)
      return 1;
if(root1 == NULL || root2 == NULL)
      return 0;
return (root1->data == root2->data && fun(root1->left, root2->left) && fun(root1->right, root2->right));
}
```

The above code describes, which of the following task?

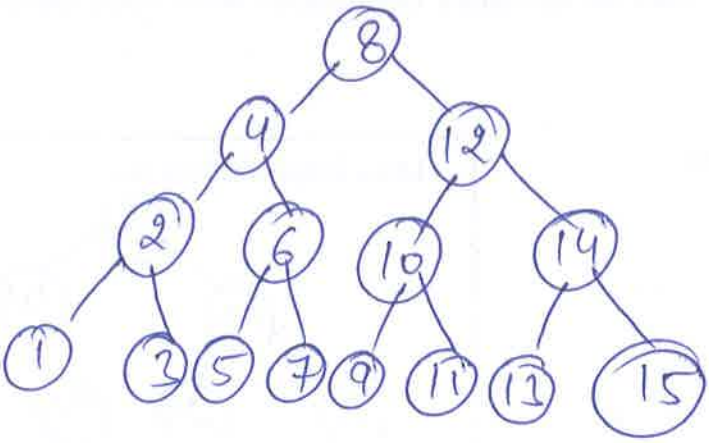
(a) Finding the number of nodes with only one child
(b) Finding the total number of nodes in a tree
(c) Finding, whether two binary trees are structurally identical or not
(d) Finding the number of leaf nodes in a tree

15. Consider the set of keys K = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 }.

(a) Draw a 2-3-4 tree storing K as its keys using the fewest number of nodes.

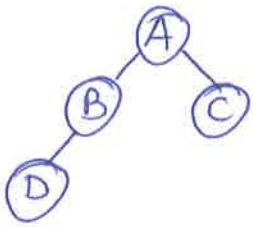


(b) Draw a 2-3-4 tree storing K as its keys using the maximum number of nodes.

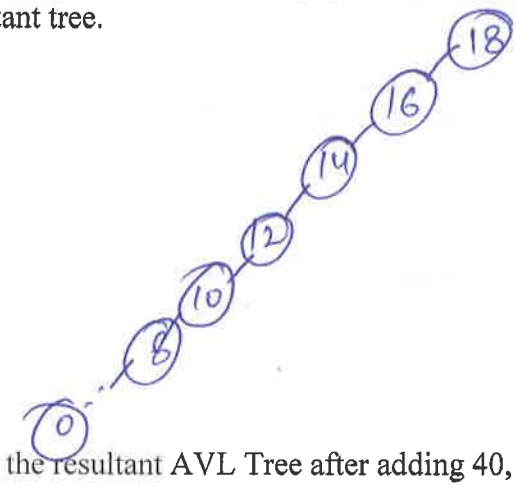


16. In a single-threaded version of the following tree, how many threads are there?

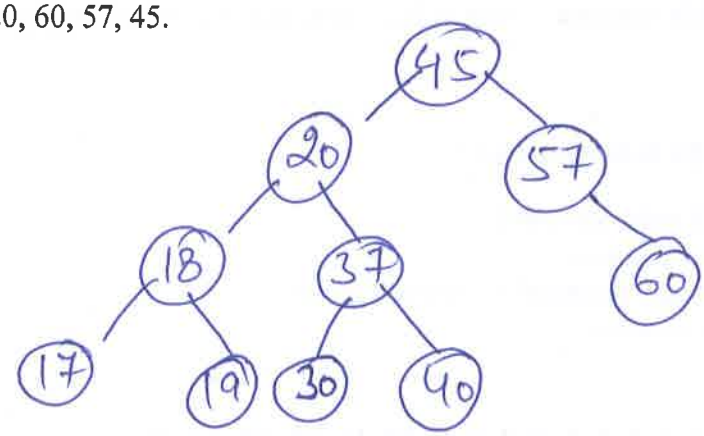
(a) 3
(b) 4
(c) 5
(d) None of these



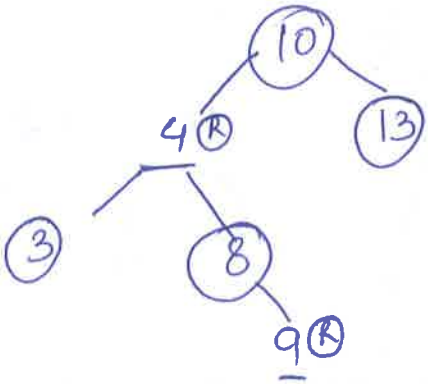
17. Insert keys 0, 2, 4, 6, 8, 10, 12, 14, 16, 18, in this order in an initially empty splay tree and draw the resultant tree.



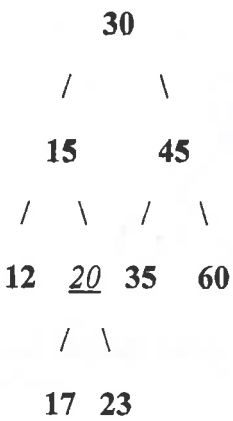
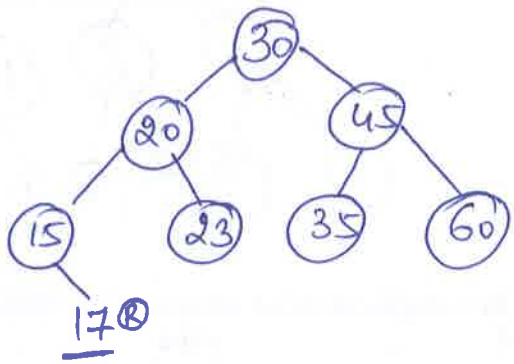
18. Draw the resultant AVL Tree after adding 40, 18, 19 to the AVL tree whose post order traversal is 17, 37, 30, 20, 60, 57, 45.



19. Construct a red-black tree by inserting the keys in the following sequence into an initially empty red-black tree: 13, 10, 8, 3, 4 and 9. Draw only the resultant tree.



20. Show the result of deleting 12 from the Red-Black tree depicted below (only underlined nodes are red):



Draw Resultant Tree here: